Knee Surg Sports Traumatol Arthrosc (2007) 15:654–664
DOI 10.1007/s00167-006-0108-8

SPORTS MEDICINE

Effect of isokinetic training on strength, functionality and proprioception in athletes with functional ankle instability

Ufuk Sekir · Yavuz Yildiz · Bulent Hazneci · Fatih Ors · Taner Aydin

Received: 6 July 2005 / Accepted: 11 January 2006 / Published online: 13 June 2006

Abstract The purpose of this study was to investigate the effects of isokinetic exercise on strength, joint position sense and functionality in recreational athletes with functional ankle instability (FAI). Strength, proprioception and balance of 24 recreational athletes with unilateral FAI were evaluated by using isokinetic muscle strength measurement, ankle joint position sense and one leg standing test. The functional ability was evaluated using five different tests. These were; single limb hopping course (SLHC), one legged and triple legged hop for distance (OLHD–TLHD), and six and cross six meter hop for time (SMHT–CSMHT). Isokinetic peak torque of the ankle invertor and evertor muscles were assessed eccentrically and concentrically at test speeds of 120°/s. Isokinetic exercise protocol was carried out at an angular velocity of 120°/s. The exercise session was repeated three times a week and lasted after 6 weeks. At baseline, concentric invertor strength was found to be significantly lower in the functionally unstable ankles compared to the opposite healthy ankles ($p < 0.001$). This difference was not present after executing the 6 weeks exercise sessions ($p > 0.05$). Ankle joint position sense in the injured ankles declined significantly from 2.35 ± 1.16 to 1.33 ± 0.62° for 10° of inversion angle ($p < 0.001$) and from 3.10 ± 2.16 to 2.19 ± 0.98° for 20° of inversion angle ($p < 0.05$) following the isokinetic exercise. One leg standing test score decreased significantly from 15.17 ± 8.50 to 11.79 ± 7.81 in the injured ankles ($p < 0.001$). Following the isokinetic exercise protocol, all of the worsened functional test scores in the injured ankles as compared to the opposite healthy ankles displayed a significant improvement ($p < 0.01$ for OLHD and CSMHT, $p < 0.001$ for SLHC, TLHD, and SMHT). These results substantiate the deficits of strength, proprioception, balance and functionality in recreational athletes with FAI. The isokinetic exercise program used in this study had a positive effect on these parameters.

Keywords Strength · Proprioception · Functional ability · Functional ankle instability · Recreational athlete

U. Sekir (✉)
Department of Sports Medicine,
Medical School of Uludag University,
16059 Gorukle, Bursa, Turkey
e-mail: ufuksek@gmail.com

Y. Yildiz · T. Aydin
Department of Sports Medicine,
Gulhane Military Academy of Medicine,
Ankara, Turkey

B. Hazneci
Department of Physical Medicine and Rehabilitation,
Gulhane Military Academy of Medicine, Ankara, Turkey

F. Ors
Department of Radiodiagnostic Radiology,
Gulhane Military Academy of Medicine, Ankara, Turkey

Introduction

Ankle injuries are one of the most common injuries encountered in sport activities. Eighty-five percent of ankle injuries are sprains, and of these, 85% are inversion sprains of the lateral ligaments [11, 36]. After an ankle sprain, 40% of sufferers continue to report residual disability involving the sensation of giving way or the feeling of instability at the ankle even when there has been no loss of passive mechanical restraints [34]. This disability is referred to as functional ankle instability (FAI) and was first introduced by Freeman et al. [8] and has been refined to mean joint motion beyond voluntary control but not necessarily exceeding physiologic (anatomical) range of motion [38].

Many authors have suggested that FAI is related to proprioceptive deficits [5, 10, 13, 19, 30, 31], mechanical instability [23] and weakness of the peroneal muscles [37]. Deficits in evertor strength would reduce the ability of these muscles to resist inversion and return the foot to a neutral position and thereby prevent inversion sprain. Recently, studies have found that concentric evertor strength deficits do not occur in FAI [27, 42]. On the contrary, other researchers have demonstrated eccentric evertor weakness in patients suffering from chronic unstable ankle compared to healthy subjects [14, 45]. Interestingly, Wilkerson et al. [42] and Ryan [33] have shown concentric invertor strength deficits in subjects with ankle sprain. They theorized that the inversion weakness could be the result of selective reflex inhibition of the ankle invertors' ability to start moving in the direction of initial injury or deep peroneal nerve dysfunction as a result of overstretching the peroneal nerve. They also speculated that the motor neuron pool associated with invertor muscle function is rendered less excitable by lateral ankle sprain, whereas that associated with evertor function is not affected to the same extent.

As formerly been mentioned, the other responsible mechanism contributing to FAI is a deficit in proprioception. Proprioception was formerly been described by Sherrington as a product of sensory information gathered to the central neural system by mechanoreceptors located in the joint-capsule, ligaments, muscles, tendons, and skin [35]. The proprioceptive mechanism is essential for proper function of the joint in sports, for activities of daily life, and for some occupational tasks. Proprioception contributes to the motor programming for neuromuscular control required for precision movements and also contributes to muscle reflex, providing dynamic joint stability. Trauma to ligamentous tissues that contain mechanoreceptors may result in partial deafferentation, which can lead to proprioceptive deficits and subsequently contributes to FAI [8, 25] Studies have shown decreased proprioceptive abilities in patients with chronically unstable ankles [8, 9, 12]. Glencross and Thornton [12] studied joint position sense and reported significantly greater errors in the functionally unstable ankle compared with the uninvolved ankle. It was also found by Konradsen and Ravn [20] that chronic ankle instability resulted in a prolonged peroneal reaction time in response to a sudden inversion stress and thereby limited the muscular control of the ankle.

Since it has been shown that repetitive ankle joint injuries cause neurosensorial, proprioceptive and mechanical impairments, exercises that increase proprioception, balance and functional capacity are routinely performed after an ankle joint injury in addition to strengthening the muscles. In order to achieve a full functional activity after ankle sprain, proprioceptive deficits and functional capacity should be restored beside the muscle strength rehabilitation intervention. An ideal rehabilitation program that is going to be carried out should firstly expose and afterwards focus on these deficits. For many years, isokinetic exercise is used to strengthen muscles after sport injuries and to increase muscle performance in healthy athletes. However, there is no study that evaluated the isokinetic exercise intervention in patients with FAI. On the other hand, studies conducted by Kaminski et al. [17] and Docherty et al. [7] employed progressive-resistance strength training in FAI. While Kaminski et al. [17] presented no significant improvements in ankle evertor and invertor muscle strength, Docherty et al. [7] reported improvements in eversion and dorsiflexion strength and in joint position measures.

The purpose of this research is to identify proprioceptive, functional, and strength deficits among a group of recreational athletes with FAI, and to determine the influence of isokinetic exercise on these parameters.

Materials and methods

Subjects

Twenty-four male recreational athletes with unilateral FAI (mean (SD) age 21 (2) years,) participated in this study. Eleven of the 24 subjects experienced symptoms of unilateral ankle instability of their dominant and thirteen on their non-dominant limb. Since previous work has demonstrated no significant strength differences for inversion and eversion movements between dominant and non-dominant limbs, the unaffected opposite ankles were used as controls [29]. Subjects were included in the study if they met the following criteria; had sustained at least two moderate sprains to the same ankle which required medical intervention and who complained of repeated episodes of "giving way", both within the last 6 months. All patients were examined by the same clinician (YY) and found to have no mechanical instability before participating. The criteria of moderate sprain was defined as having moderate pain, swelling, and tenderness over the involved ligaments without a significant instability (a definite end point was present on ligamentous testing). No subjects had suffered injury to the contralateral healthy ankles for at least 6 months before testing, and none of them were undergoing rehabilitation of the injured ankle before initiating the test procedure, nor had any complaints of pain, swelling, or functional limitations in the injured ankles during the test period. Considering the opposite healthy ankle joint, active range of motion (dorsiflexion/plantarflexion; inversion/eversion) as measured using an electronic goniometer (Cybex, EDI 320, AL, USA) was found to be within normal limits in the injured ankle joint for all subjects. No subjects were involved in physical activity that exceeded three sessions a week for more than half an hour per session. Written consent was obtained from each subject before testing, and all subjects were screened to ensure that there were no lower extremity neuromuscular or musculoskeletal problems or contraindications for isokinetic testing. After being informed about the study and test procedures, and any possible risks and discomfort that might ensue, their written informed consent to participate was obtained in accordance with the Helsinki Declaration [44].

Experimental procedure

Before and after the isokinetic exercise treatment all the testing protocol was carried out on two separate days. Isokinetic strength and functional test measurements were made on the first day. Thereafter, on the second day, balance and proprioceptive ability of the ankle joint was evaluated.

Isokinetic strength measurement

Isokinetic testing of the ankle invertor and evertor muscles was performed at a velocity of 120°/s for eccentric and concentric contractions of the injured and uninjured ankles. Subjects performed a ten-minute warm-up of general range of motion and stretching exercise for joint movements of inversion/eversion and dorsiflexion/plantarflexion. After the warm-up, they were appropriately positioned on the isokinetic dynamometer (Cybex), which was calibrated before testing each subject (Fig. 1). The ankle was positioned in order that the subtalar joint became in neutral, which was the start position (0°) [32]. The talocrural joint was positioned in 10–15° plantarflexion as a consequence of the low cut lace up shoe worn by each subject that simulated a position for inversion injury. Two straps, which crossed to the dorsum of the foot, were attached to the footplate. The knee of the test leg was positioned in 80–110° flexion and the lower leg was parallel to the floor. The thigh stabilizer pad and strap secured the distal aspect of the thigh for the test leg and a seatbelt placed around the abdomen secured the torso. All the tests were performed with the subjects wearing shoes.

The range of motion stop angles from eversion to inversion direction were set at 20° of eversion and 30° of inversion, and vice versa from inversion to eversion direction. Three sub-maximal trials for familiarizing with the isokinetic testing procedure and to perform warm-up were followed by five maximal concentric and eccentric invertor and evertor trials. Evertor and invertor muscle strength was obtained by measuring maximal force moments (torque) during isokinetic ankle inversion and eversion movements. To ensure that a maximal effort was attained, all subjects received positive verbal encouragement during testing. The same investigator performed all the tests to ensure standardization. A two-minute rest was permitted between the test for inversion and eversion to prevent the build up of fatigue. The Intraclass correlation coefficients (ICCs) for ankle isokinetic strength measurements at 120°/s varied from 0.86 to 0.89 (see data analysis).

Proprioceptive ability of the ankle joint

In order to evaluate the proprioceptive ability of the ankle joint the measurement of ankle joint position sense was used.

Ankle joint position sensibility

Passive joint position sense (PJPS) was measured by the continuous passive motion (CPM) mode of the

Fig. 1 Position of the subject on the Cybex Norm isokinetic dynamometer for ankle eversion and inversion strength testing

isokinetic dynamometer (Cybex NORM™) at 1°/s angular speed. The tested foot was placed on the footplate of the Cybex, according to the manufacturer's instructions for isolating inversion-eversion and plantar flexion-dorsiflexion, and was secured with Velcro straps. For this study, like strength measurements, the uninjured ankles served as the control limb for all tests since the proprioception of the lower extremity did not appear to be influenced by limb-dominance [1]. Prior to testing, the Cybex dynamometer was calibrated as a part of the regular schedule for maintenance of equipment used for this testing device. The testing order of the injured and uninjured limbs was randomized prior to testing.

To initiate the test, the foot was placed in the neutral (0°) position. All subjects were blindfolded to eliminate the contribution of visual cues during the repositioning of the joint. For familiarizing with the testing device, subjects were instructed to perform three active repetitions of ankle movement ranging from maximal ankle inversion to maximal eversion. The test began with the tester passively moving the tested ankle into the testing position of 10° of inversion and maintaining that position for ten seconds. After ten seconds of static positioning, the ankle was moved back passively from the presented angle to the reference angle (neutral position). When the testing device moved continuously from the neutral position to inversion at 1°/s angular speed the subject was asked to passively reproduce the previously presented test angle of 10° of inversion by stopping the device using the handheld on-off switch as he thought the test angle had been reached. Two trials were performed. Following the first reference angle, the same testing protocol was used for 20° of inversion angle from the beginning of the starting angle (neutral position). Angular displacement was recorded as the error in degrees between the reference angle and the repositioned angle. The mean of the two trials for each tested condition was calculated, to determine an average error in scores [1]. The ICCs for 10 and 20° of inversion were 0.90 and 0.94, respectively (see data analysis).

One leg standing test

This test evaluates the subject's ability to keep balance while standing on one leg. We performed this test not on a hard surface but on a medium-density polyfoam mat and with eyes closed to increase the failure rate. The subjects stood on the test side limb with their stance foot centered on the mat while their knees are slightly flexed. They were instructed to lift the limb that was not being tested by bending the knee, and holding it in ~90° of knee flexion. Once the subjects were in this position with eyes closed, and said that they were ready, data collection was started. The one leg standing test measurement was performed for one minute. During the test period, each surface contact with the contralateral leg, moving the test foot, or swaying the body excessively out from midline in any direction to obtain a balanced stance was counted as one failure point. The subjects performed the tests without shoes and socks to negate any extraneous skin sensation from clothing touching the foot area. The outcome measure was averaged over two trials. The ICC for the one leg standing test was 0.92 (see data analysis).

Functional ability of the ankle joint

We evaluated the functional ability of the ankle using five different tests. The tests that performed were; the single limb hopping course (SLHC), the one legged hop for distance, the triple legged hop for distance (OLHD–TLHD), the six meter hop for times (SMHTs) and the cross six meter hop for times (CSMHTs).

Single limb hopping course

This test is especially useful to document the function of the ankle on an uneven surface and was previously described by Aydin et al. [1] The jumping course consists of eight squares, four of them are even, and, one square has a 15° increase, another square has a 15° decrease, and two squares show a 15° lateral inclination (Fig. 2). The volunteers are asked to jump across this course on one leg by touching each area once as fast as possible without leaving the course. The test result is quantified by seconds used to pass the course. Each failure adds an extra second to the time taken to complete the course. The ICC for this test was 0.96 (see data analysis).

One legged and triple legged hop for distance

Patients were asked to make one and three forceful hopping movements forward as large as they could. The distance between the starting point and the end point was measured. Two tests were performed and the average distance was measured for each test. The ICCs for OLHD–TLHD were 0.97 and 0.98, respectively (see data analysis).

Six meter and cross six meter hop for time

This is a timed test performed over a distance of 6 m. Each subject was encouraged to use linear, large,

Fig. 2 Single limb hopping course

forceful one legged hopping motions and crosswise, large, forceful one legged hopping motions across a line with a 10 cm width to propel his body toward the measured distance as quickly as possible. Two tests were performed and the average time was measured for each test. The ICCs for SMHT and CSMHT were 0.91 and 0.89, respectively (see data analysis).

Isokinetic exercise protocol

The Cybex Norm isokinetic system was used for the isokinetic exercise program. Only the injured ankles were taken into the exercise sessions. Since there were no strength deficits found for the eccentric mode of the invertor and evertor muscles from the initial strength measurements, the isokinetic exercise program of the ankle joint was performed only in the concentric mode for both the inversion and eversion movements. Each exercise session was carried out with three settings of 15 repetitions at 120°/s. These sessions were repeated three times a week and lasted after 6 weeks.

Data analysis

The level of significance was set at $p < 0.05$. Statistical analysis was performed using SPSS version 10.0 (SPSS, SPSS Inc, Chicago, IL, USA) software. All tests were two-tailed.

Statistical differences between the normally distributed data (i.e. ankle muscle strength parameters, JPS at 10°, OLST, OLHD, TLHD, and SMHT) within the uninjured and injured ankle groups before and after intervention were investigated using paired *t*-tests, and differences between groups using unpaired *t*-tests. Within group differences between data that were skewed (i.e. JPS at 20°, SLHC, and CSMHT) were investigated using Wilcoxon's signed rank tests, and between group differences using the Mann–Whitney *U*-test.

The Pearson product-moment correlation test was performed among the percentage changes in isokinetic ankle evertor and invertor muscle strength, proprioception values and functional tests.

Ankle isokinetic strength, ankle joint position sensibility, one leg standing test and the five different functional ability measurements were repeated twice at 3–5 days intervals for reliability analysis in 20 healthy subjects. The ICC were used to determine the reliability of these tests. The ICC was accepted as clinically meaningful if values were equal or greater than 0.80.

Results

Isokinetic strength of ankle evertor and invertor muscles

Mean peak torque values of the ankle evertor and invertor muscles before and after the training are presented in Table 1. As shown in the table, only the concentric invertor peak torque was lower in the injured ankles at pre-training ($p = 0.000001$) Although the concentric torque values of the evertors were lower in the injured ankles, no significant difference was observed between the injured and non-injured ankles at baseline ($p > 0.05$). Nevertheless, concentric evertor muscle strength exhibited a statistically significant increase after exercise as to the pre-exercise level in the injured ankle group ($p = 0.013$). Similarly, the concentric invertor strength of the injured ankle group showed a significant increase after exercise ($p = 0.007$).

Proprioceptive ability and balance

At pre-treatment evaluation passive reproduction error score of ankle joint position sense was 2.35 ± 1.16° at 10° of inversion angle and 3.10 ± 2.16° at 20° of inversion angle in the injured ankles and were significantly higher as to the non-injured ankles ($p = 0.006$ for 10° of inversion angle, $p = 0.026$ for 20° of inversion angle). These values represented a significant improvement following exercise and were 1.33 ± 0.62° for 10° of inversion ($p = 0.001$) and 2.19 ± 0.98° for 20° of inversion ($p = 0.019$). Furthermore, the worsened

Table 1 Muscle strength values in the injured and uninjured ankles before and after the exercise intervention (Mean ± SD)

	Injured ankle		Uninjured ankle	
	Pre-exercise	Post-exercise	Pre-exercise	Post-exercise
EccEvertorPT (Nm)	25.17 ± 2.30	26.17 ± 3.78	26.25 ± 2.89	26.96 ± 3.61
EccInvertorPT (Nm)	27.42 ± 5.26	28.17 ± 5.09	29.33 ± 5.47	28.96 ± 4.76
ConEvertorPT (Nm)	15.54 ± 3.13	17.21 ± 3.36**	16.96 ± 2.68	17.38 ± 3.56
ConInvertorPT (Nm)	14.79 ± 1.56*	16.50 ± 2.38***	17.46 ± 1.59	16.92 ± 2.12

Ecc Eccentric, *Con* Concentric, *PT* Peak torque, *Nm* Newton-meter

*$p = 0.000001$, $t = 5.870$ (between groups); **$p = 0.013$, $t = 2.681$ (following exercise); ***$p = 0.007$, $t = 2.984$ (following exercise)

one leg standing test score in the injured ankles compared to the opposite ankles at pre-treatment ($p = 0.001$) showed also a significant decrease from 15.17 ± 8.50 to 11.79 ± 7.81 failure point after the intervention ($p = 0.001$). Besides these improvements in the injured ankle group, also the one leg standing test score in the uninjured ankle group represented a significant decrease from 8.44 ± 4.09 to 7.88 ± 3.77 failure point ($p = 0.008$) (Table 2).

Functional performance

Before intervention, the time taken to perform the three individual tests of functional performance (SLHC, SMHT, and CSMHT) was significantly prolonged and the distance in the OLHD was significantly shortened in the injured ankle group compared to the uninjured ankle group ($p = 0.029$ for SLHC; $p = 0.003$ for OLHD; $p = 0.030$ for SMHT; $p = 0.001$ for CSMHT; Table 3). There was no change in the functional tests of the uninjured control ankles, but following exercise, all the five functional performance test parameters of the injured ankle group displayed an improvement ($p = 0.0001$ for SLHC; $p = 0.003$ for OLHD; $p = 0.001$ TLHD; $p = 0.001$ for SMHT, and $p = 0.01$ for CSMHT; Table 3).

Correlation analysis

The isokinetic data did not correlate with the proprioceptive measures and functional parameters in the unstable ankles following training (Table 4).

Discussion

The main purpose of this study was to explore the effects of an isokinetic exercise intervention of 6 weeks duration on muscle strength, proprioception, balance, and functional capacity. In recent years, isokinetic exercises have been commonly used in sports medicine practice and have increased muscle conditioning and reduced clinical complaints [46]. It was recently shown by Hazneci et al. [15] that besides the improvements in strength, also the proprioceptive ability is increasing after a muscular rehabilitation program carried out isokinetically in patients with patellofemoral stress syndrome. There are specific exercise programs to improve muscle strength, proprioception and functional capacity of the ankle. However, an ideal exercise program should improve not only the neuromuscular stability but also the functional capacity. To our knowledge, the effects of isokinetic exercises on functional capacity, proprioceptive ability, and muscular

Table 2 Joint position sense and one leg standing test scores in the exercise performed (injured) and not performed (healthy) ankles (Mean ± SD)

	Injured ankle		Uninjured ankle	
	Pre-exercise	Post-exercise	Pre-exercise	Post-exercise
AJPS10-In (degree)	2.35 ± 1.16*	1.33 ± 0.62****	1.50 ± 0.86	1.38 ± 0.73
AJPS20-In (degree)	3.10 ± 2.16**	2.19 ± 0.98*****	1.79 ± 1.06	1.88 ± 0.96
OLST	15.17 ± 8.50***	11.79 ± 7.81******	8.44 ± 4.09	7.88 ± 3.77*******

AJPS10-In Ankle joint position sense at 10° of inversion angle, *AJPS20-In* Ankle joint position sense at 20° of inversion angle, *OLST* One leg standing test

*$p = 0.006$, $t = 2.905$ (between groups); **$p = 0.026$, $z = 2.219$ (between groups); ***$p = 0.001$, $t = 3.494$ (between groups); ****$p = 0.001$, $t = 3.647$ (following exercise); *****$p = 0.019$, $z = 2.348$ (following exercise); ******$p = 0.001$, $t = 4.127$ (following exercise); *******$p = 0.008$, $t = 2.908$ (following exercise)

Table 3 Effects of isokinetic exercise on functional test scores in recreational athletes having unilateral FAI (Mean ± SD)

	Injured ankle		Uninjured ankle	
	Pre-exercise	Post-exercise	Pre-exercise	Post-exercise
SLHC (sec)	13.27 ± 6.27*	10.44 ± 3.56*****	9.58 ± 2.98	9.29 ± 2.35
OLHD (cm)	105.79 ± 40.33**	116.88 ± 39.88******	135.21 ± 21.66	135.58 ± 21.03
TLHD (cm)	365.96 ± 135.22	399.96 ± 137.10*******	427.79 ± 116.62	429.54 ± 115.69
SMHT (sec)	3.93 ± 1.84***	3.23 ± 1.40********	2.98 ± 0.99	2.90 ± 0.92
CSMHT (sec)	3.38 ± 1.22****	2.84 ± 1.08*********	2.35 ± 0.62	2.29 ± 0.61

SLHC Single limb hopping course, *OLHD–TLHD* One legged and triple legged hop for distance, *SMHT–CSMHT* Six meter and cross six-meter hop for time

*$p = 0.029$, $z = 2.190$ (between groups); **$p = 0.003$, $t = 3.148$ (between groups); ***$p = 0.030$, $t = 2.236$ (between groups); ****$p = 0.001$, $z = 3.268$ (between groups); *****$p = 0.0001$, $z = 4.022$ (following exercise); ******$p = 0.003$, $t = 3.366$ (following exercise); *******$p = 0.001$, $t = 3.779$ (following exercise); ********$p = 0.001$, $t = 3.651$ (following exercise); *********$p = 0.01$, $z = 3.225$ (following exercise)

Table 4 Pearson product-moment correlation coefficients (r) among percentage change values of the isokinetic parameters and proprioception and functional performance measures in the unstable ankles following training ($n = 24$)

	AJPS10-In	AJPS20-In	OLST	SLHC	OLHD	TLHD	SMHT	CSMHT
EccEvertorPT	–.047	.091	.213	–.046	–.054	–.142	–.207	.043
EccInvertorPT	–.073	.124	.055	–.201	–.062	–.266	–.239	.083
ConEvertorPT	.020	.273	–.221	–.064	.293	.221	.016	.207
ConInvertorPT	–.005	.141	.126	.226	–.059	–.247	–.311	–.038

strength in subjects with functional ankle sprain have not been studied previously. There are two studies that investigated the effects of either strength and proprioception training in combination [17] or strength training alone [7] on strength and proprioceptive parameters in functionally unstable ankles. However, these studies utilized a progressive resistance protocol using Thera–Band elasticated bands, instead of an isokinetic intervention. Kaminski et al [17] concluded that 6 weeks of strength and proprioception training had no effect on strength in subjects with unilateral functional instability. On the other hand, Docherty et al [7] reported improvements in eversion and dorsiflexion strength after 6 weeks of progressive resistance strength training in unilateral FAI. Their subjects have also shown improvements in joint position sense measures, a finding the researchers attributed to enhancements in muscle spindle activity.

Kaminski et al. [18] suggested that, unless evertor weakness is demonstrated, evertor strengthening exercises for patients with FAI would not seem be warranted. Since we found strength deficits only in the concentric invertor muscles during the initial strength measurements, the isokinetic exercise program of the ankle joint was performed only in the concentric mode in the present study both for the invertor and evertor muscles. The reasons for including the evertor muscles to the isokinetic exercise program were; firstly, although not significantly different between the ankles, the concentric torque values of the evertors were also lower in the unstable ankle and secondly, we thought while the isokinetic testing device moves continuously in the concentric mode from maximal eversion to maximal inversion, the subjects could have problems about adapting to only perform inversion exercise through this range and they could not produce their maximal effort. We used the same testing and training isokinetic speeds (120°/s angular velocity) in the present study. In this way we could evaluate small significant clinical improvements before and after the isokinetic training.

The present study exhibited significant improvements in strength, proprioception and functional performance. However, the correlation analyses showed no relation between the improvements of these parameters (Table 4). This indicates that there is not necessarily a cause-effect relationship between each of these variables.

Ankle strength: It is thought that the peroneal muscles must be strong enough to counter the inversion mechanism associated with FAI [4, 37]. Several investigations, however, have contradicted the premise that peroneal muscle weakness is associated with chronic ankle instability [3, 8, 23, 24, 33, 42]. As in these studies, our study also did not show evertor muscle weakness in subjects with unilateral FAI compared to the opposite healthy ankle. Few studies have investigated eccentric evertor and invertor ankle strength in FAI. Up to now, four studies have been compared the injured limb with the non-injured in subjects with FAI

[2, 16, 18, 29]. None of these studies have shown a deficit in eccentric evertor muscle strength. Consistent with these studies the present study did also not display eccentric evertor strength deficits in subjects having FAI. On the contrary, other studies have established eccentric evertor muscle weakness in patients with FAI compared to healthy individuals [14, 43, 45].

Also interestingly, the current study has demonstrated concentric invertor strength deficits in functional unstable ankles. Previous studies support these results [27, 33, 42]. Ryan [33] theorized that the inversion deficits might have resulted from selective inhibition or deep peroneal nerve dysfunction as a result of overstretching the peroneal nerve. Wilkerson et al. [42] suggested that, despite the widespread focus on strengthening of the evertors in ankle rehabilitation, the latest evidence suggests a relationship between deficits in invertor strength and lateral ligament injury. On the other hand, other recent studies have not been demonstrated concentric invertor muscle weakness in the disabled limbs compared with contralateral normal limbs or an age-matched healthy group [29, 45] Although there are confusing results, one thing that is definite is that strength deficits of the invertor muscles are existed in functionally unstable ankles (Table 5). We hypothesized that ankle invertor strength might play a role in preventing loss of lateral postural stability over a fixed foot, which in turn may result in excess ankle inversion [41, 42] Reduced invertor strength may be reflected by muscle imbalance between the evertor and invertor muscle groups about the ankle.

As previously mentioned only the concentric invertor muscle strength was significantly lower on the functional unstable ankle than on the opposite healthy ankle at pre-treatment level. Both the concentric invertor strength and, although not significantly different between the ankles at baseline, the concentric evertor strength displayed also a significantly increase following the isokinetic exercise intervention. These results confirm the effectiveness of isokinetic training on gaining strength in functional unstable ankles.

Proprioception and balance: Proprioception is the cumulative neural input to the central neural system from mechanoreceptors located in the joint-capsule, ligaments, muscles, tendons, and skin. These me-

Table 5 Studies concerning ankle invertor and evertor strength in FAI

Authors	Test speed	Subject group	Results
Lentell et al. [24]	0 and 30°/s	Unilateral CAI and uninjured opposite ankles	No difference in concentric evertor strength between ankles
Lentell et al. [23]	30, 90, 150, and 210°/s	Unilateral FAI and uninjured opposite ankles	No difference in eversion strength between ankles
Kaminski et al. [18]	Seven different angular velocities	Subjects with CAI and healthy control group	No difference in concentric and eccentric evertor strength between groups
Bernier et al. [2]	90°/s	Unilateral FAI and uninjured opposite ankles (n = 15)	No difference in concentric and eccentric invertor and evertor strength between ankles
Heitman et al. [16]	30 and 60°/s	Unilateral FAI and uninjured opposite ankles (n = 9)	No difference in eccentric evertor strength between ankles
Munn et al. [29]	60 and 90°/s	Unilateral FAI and uninjured opposite ankles (n = 16)	No difference in eccentric or concentric evertor strength between ankles
Hartsell and Spaulding [14]	60, 120, 180, and 240°/s	Subjects with FAI (n = 14) and healthy control group (n = 10)	Significant difference in eccentric and concentric invertor and evertor strength between groups
Willems et al. [43]	30 and 120°/s	Subjects with CAI (n = 10) and healthy control group (n = 53)	Significant difference in eccentric and concentric evertor strength with no difference in concentric strength between groups
Yildiz et al. [45]	120°/s	Subjects with CAI (n = 8) and healthy control group (n = 9)	Significant difference in eccentric evertor strength between groups
Ryan [33]	30°/s	Unilateral CAI and uninjured opposite ankles (n = 45)	No difference in concentric evertor strength, significant difference in concentric invertor strength between ankles
Wilkerson et al. [42]	30 and 120°/s	Unilateral CAI and uninjured opposite ankles (n = 15)	No difference in evertor strength, significant difference in invertor strength between ankles

chanoreceptors are sensitive to joint pressure and tension caused by dynamic movement and static position [2]. This disturbance of the afferent input from mechanoreceptors would affect not only a sense of movement and position but also a subsequent proprioceptive reflex to control posture and coordination. Therefore, functional instability of the ankle has been studied using tests of joint position sense [2, 31], reaction time of the peroneal muscles in a simulated ankle sprain [21, 36], and postural sway while standing on a single limb [2, 13, 22, 30]. Consequently, these parameters are significantly useful in investigating functionally unstable ankles and have been reported to be associated with the symptoms of FAI as well [5, 10] A considerable amount of attention has been given to examining deficits in proprioception as a cause of FAI. Several researchers [8, 12, 24, 38] have concluded that long-term ankle instabilities can be related to decreased joint proprioception. In fact, several recent reports [23, 26] have suggested that decreased proprioception as a cause of FAI is more important consideration than first thought. In the present study, the proprioceptive ability and balance was tested using the measurement of ankle joint position sense and one leg standing test. Both measurements displayed deficits in patients having FAI. Abnormal proprioceptive awareness was found at 10° of inversion and 20° of inversion, suggesting that the results are not confined to a specific part of range. Ankle joint position sense and one leg standing test values attained similar levels as the uninjured control ankles after the isokinetic exercise. Regarding these results we can say that the proprioceptive ability and balance could be recuperated by the isokinetic exercise used in this study. The improvements of the joint position sense scores after gaining strength in the muscles might be resulted due to two reasons: first one is, the imbalance between the invertor and evertor muscle strength may led to biomechanical imbalance of the ankle joint and subsequently could result in stimulation of the nociceptors. After increasing strength this might have balanced the biomechanics of the ankle and led to the disappearance of the nociceptor stimuli, consequently the proprioceptive stimuli that is transported to the central nervous system via the group A beta fibers might be enhanced. The second reason might be due to the enhancements in muscle spindle and Golgi tendon organ activity as reported also by Docherty et al. [7] As it is known, when a joint moves, impulses must arise from several classes of afferent for providing proprioceptive signals: in addition to input from afferents arising from the ligament and joint capsule, there is also input from proprioceptive receptors located in the cutaneous, muscle (muscle spindle) and tendon (golgi tendon organ) tissues. After strengthening the muscle structures this could be enhanced the proprioceptive abilities via stimulating the muscle spindle and golgi tendon organ receptors. The muscle spindles receive stimuli from static and dynamic gamma efferent nerves, and it is possible that strength training might have increased the gamma efferent activity, thereby resulting in greater acuity in sensing joint position.

Functional performance: Proprioceptive damage caused by an ankle sprain (secondary alteration) may impair the feedback needed to retain well-functioning central motor programs. The injury might directly alter the motor programs so that abnormal, injury-related motor performance occurs when an ankle is perturbed. While researchers have demonstrated that the afferent feedback system may be interrupted after injury, knowledge of how these deficits may affect actual performance on functional tasks is scarce. In the few studies to date, functional-performance measures such as the shuttle run and various single-leg hopping tasks have been examined, and no significant difference in any of the functional-performance tasks was found when comparing the injured with the uninjured side [28]. Differently, Demeritt et al. [6] have investigated three functional-performance tests (cocontraction, shuttle run, and agility hop test) in patients with instability and in healthy groups. They also could not demonstrate any differences between the groups and suggested that FAI do not negatively influence actual functional performance. In contrary to these results, the present study has shown worsened functional test scores in the functional unstable ankles compared to the contralateral ankles. Moreover, after the isokinetic exercise intervention the functional capacity of the injured ankles reached the same level as the uninjured ankles. These shows that functional performance is negatively affected in patients with FAI and that a strengthening exercise performed isokinetically could ameliorate functional performance. Interestingly, one leg testing score improved also in the uninjured ankles. Although cross-training effects of strength training are well known according to our knowledge, isokinetic crossover training effects have not been clearly established [39, 40, 46]. This observation is intriguing, and further studies specifically designed to investigate potential isokinetic cross-training effects are warranted. If isokinetic cross-training effects can be reliably established, rehabilitation protocols could be altered to include early isokinetic training of the contralateral ankle in cases which severe sprains preclude timely progression to strength training on the injured side. This, therefore, shows that isokinetic exercise programs can be used to increase the performance of healthy ankles, as they have positive effects on complex movement patterns.

Conclusion

The current study supports that neuromuscular control and functional capacity of the unilaterally functional unstable ankles were significant different from those of the contralateral healthy ankles. On the contrary to be thought of eccentric evertor muscle strength deficits may be present in functional unstable ankles, only concentric invertor muscle strength deficits were present in the functionally unstable ankles. The isokinetic exercise program used in this study had a positive effect on the functional ability, muscle strength and proprioception of the ankle.

References

1. Aydin T, Yildiz Y, Yildiz C, Atesalp S, Kalyon TA (2002) Proprioception of the ankle: a comparison between female teenaged gymnasts and controls. Foot Ankle Int 23:123–129
2. Bernier JN, Perrin DH (1998) Effect of coordination training on proprioception of the functionally unstable ankle. J Orthop Sports Phys Ther 27:264–275
3. Bernier JN, Perrin DH, Rijke AM (1997) Effect of unilateral functional instability of the ankle on postural sway and inversion and eversion strength. J Athl Train 32:226–232
4. Bosien WR, Staples S, Russel SW (1955) Residual disability following acute ankle sprains. J Bone Joint Surg Am 37:1237–1243
5. Brunt D, Andersen JC, Huntsman B, et al. (1992) Postural responses to lateral perturbation in healthy subjects and ankle sprain patients. Med Sci Sports Exerc 24:171–176
6. Demeritt KM, Shultz SJ, Docherty CL, Gansneder BM, Perrin DH (2002) Chronic ankle instability does not affect lower extremity functional performance. J Athl Train 37:507–511
7. Docherty CL, Moore JH, Arnold BL (1998) Effects of strength training on strength development and joint position sense in functionally unstable ankles. J Athl Train 33:310–314
8. Freeman MA, Dean MR, Hanham IW (1965) The etiology and prevention of functional instability of the foot. J Bone Joint Surg Br 47:678–685
9. Fu AS, Hui-Chan CW (2005) Ankle joint proprioception and postural control in basketball players with bilateral ankle sprains. Am J Sports Med 33:1174–1182
10. Garn SN, Newton RA (1988) Kinesthetic awareness in subjects with multiple ankle sprains. Phys Ther 68:1667–1671
11. Garrick J, Requa R (1988) The epidemiology of foot and ankle injuries in sport. Clin Sports Med 7:29–36
12. Glencross D, Thornton E (1981) Position sense following joint injury. J Sports Med Phys Fitness 21:23–27
13. Guskiewicz KM, Perrin DH (1996) Effect of orthotics on postural sway following inversion ankle sprain. J Orthop Sports Phys Ther 23:326–331
14. Hartsell HD, Spaulding SJ (1999) Eccentric/concentric ratios at selected velocities for the invertor and evertor muscles of the chronically unstable ankle. Br J Sports Med 33:255–258
15. Hazneci B, Yildiz Y, Sekir U, Aydin T, Kalyon TA (2005) Efficacy of isokinetic exercise on joint position sense and muscle strength in patellofemoral pain syndrome. Am J Phys Med Rehabil 84:521–527
16. Heitman RJ, Kovaleski J, Gurchiek L (1997) Isokinetic eccentric strength of the ankle evertors after injury. Percept Mot Skills 84:258
17. Kaminski TW, Buckley BD, Powers ME, Hubbard TJ, Ortiz C (2003) Effect of strength and proprioception training on eversion to inversion strength ratios in subjects with unilateral functional ankle instability. Br J Sports Med 37:410–415
18. Kaminski T, Perrin D, Gansneder B (1999) Eversion strength analysis of uninjured and functionally unstable ankles. J Athl Train 34:239–245
19. Konradsen L, Olesen S, Hansen HM (1998) Ankle sensorimotor control and eversion strength after acute ankle inversion injuries. Am J Sports Med 26:72–77
20. Konradsen L, Ravn JB (1990) Ankle instability caused by prolonged peroneal reaction time. Acta Orthop Scand 61:388–390
21. Konradsen L, Ravn JB (1991) Prolonged peroneal reaction time in ankle instability. Int J Sports Med 12:290–292
22. Leanderson J, Ekstam S, Salomonsson C (1996) Taping of the ankle: the effect on postural sway during perturbation, before and after a training session. Knee Surg Sports Traumatol Arthrosc 4:53–56
23. Lentell G, Baas B, Lopez D, et al. (1995) The contributions of proprioceptive deficits, muscle function, and anatomic laxity to functional instability of the ankle. J Orthop Sports Phys Ther 21:206–215
24. Lentell GL, Katzman LL, Walters MR (1990) The relationship between muscle function and ankle stability. J Orthop Sports Phys Ther 11:605–611
25. Lephart SM, Pincivero DM, Giraldo JL, Fu FH (1997) The role of proprioception in the management and rehabilitation of athletic injuries. Am J Sports Med 25:130–137
26. Lofvenberg R, Karrholm J, Sundelin G, Ahlgren O (1995) Prolonged reaction time in patients with chronic lateral instability of the ankle. Am J Sports Med 23:414–417
27. McKnight C, Armstrong C (1997) The role of ankle strength in functional ankle instability. J Sport Rehabil 6:21–29
28. Munn J, Beard D, Refshauge K, Lee RJ (2002) Do functional-performance tests detect impairment in subjects with ankle instability? J Sport Rehabil 11:40–50
29. Munn J, Beard DJ, Refshauge KM, et al. (2003) Eccentric muscle strength in functional ankle instability. Med Sci Sports Ecerc 35:245–250
30. Pintsaar A, Brynhildsen J, Tropp H (1996) Postural corrections after standardised perturbations of single limb stance: effect of training and orthotic devices in patients with ankle instability. Br J Sports Med 30:151–155
31. Robbins S, Waked E, Rappel R (1995) Ankle taping improves proprioception before and after exercise in young men. Br J Sports Med 29:242–247
32. Ronkonkoma NY (1995) Cybex Norm Int Inc: testing and rehabilitation system: pattern selection and set up: automated protocols. User's guide. Blue Sky Software Corporation, New York
33. Ryan L (1994) Mechanical stability, muscle strength, and proprioception in the functionally unstable ankle. Aust J Physiother 40:41–47
34. Safran MR, Benedetti RS, Bartolozzi AR, et al. (1999) Lateral ankle sprains: a comprehensive review. Part:1 etiol-

ogy, pathoanatomy, histopathogenesis, and diagnosis. Med Sci Sports Exerc 31:S429–S437
35. Sherrington CS (1906) On the proprioceptive system, especially in its reflex aspect. Brain 29:467–482
36. Sheth P, Yu B, Laskowski ER, et al. (1997) Ankle disk training influences reaction times of selected muscles in a simulated ankle sprain. Am J Sports Med 25:538–543
37. Tropp H (1986) Pronator muscle weakness in functional instability of the ankle joint. Int J Sports Med 7:291–294
38. Tropp H, Odenrick P, Gillquist J (1985) Stabilometry recordings in functional and mechanical instability of the ankle joint. Int J Sports Med 6:180–182
39. Weir JP, Housh DJ, Housh TJ, et al. (1997) The effect of unilateral concentric weight training and detraining on joint angle specificity, cross-training, and the bilateral deficit. J Orthop Sports Phys Ther 25:264–270
40. Weir JP, Housh TJ, Weir LL (1994) Electromyographic evaluation of joint angle specificity and cross-training after isometric training. J Appl Physiol 77:197–201
41. Wilkerson G, Nitz A (1994) Dynamic ankle stability: mechanical and neuromuscular interrelationships. J Sport Rehabil 3:43–57
42. Wilkerson GB, Pinerola JJ, Caturano RW (1997) Invertor vs. evertor peak torque and power deficiencies associated with lateral ankle ligament injury. J Orthop Sports Phys Ther 26:78–86
43. Willems T, Witvrouw E, Verstuyft J, Vaes P, De Clercq D (2002) Proprioception and muscle strength in subjects with a history of ankle sprains and chronic instability. J Athl Train 37:487–493
44. WMADH (2000) World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects. J Am Med Assoc 284:3043–3045
45. Yildiz Y, Aydin T, Sekir U, et al. (2003a) Peak and end range eccentric evertor/concentric invertor muscle strength ratios in chronically unstable ankles: comparison with healthy individuals. J Sports Sci Med 2:70–76
46. Yildiz Y, Aydin T, Sekir U, et al. (2003b) Relation between isokinetic muscle strength and functional capacity in recereational athletes with chondromalacia patellae. Br J Sports Med 37:475–479